

Class 10 Science - Chapter 13

Our Environment (Hinglish Notes)

PART A - CORE BASICS (MISS MAT KARNA)

Class 7-8 ki yeh cheezein pehle pakki karo, warna Ch13 samajh nahi aayega.

1. Environment aur Organisms (Class 7)

Environment (paryavaran) = jeev ke charo taraf ki saari cheezein (hawa, paani, mitti, doosre jeev).

Biotic = jeevit cheezein (paudhe, jaanwar, bacteria).

Abiotic = nirjeev cheezein (hawa, paani, dhoop, mitti, temperature).

2. Nutrition types (Ch5 link)

Autotroph (producer) = khud bhojan banaye - green plants (photosynthesis).

Heterotroph (consumer) = doosron ka khaye - jaanwar.

Decomposer = mre jeev/waste ko todkar khaye - bacteria aur fungi.

3. Energy aur Food (Class 7-8)

Energy ka asli source = SURAJ (sun).

Producer suraj ki energy ko food (chemical energy) me badalte.

Food chain = "kaun kisko khata" ka kram.

4. Photosynthesis aur Respiration (revision)

Photosynthesis: $\text{CO}_2 + \text{paani} + \text{sunlight} \rightarrow \text{glucose} + \text{O}_2$ (producer karte).

Respiration: glucose tootkar energy (saare jeev karte).

Energy ek roop se doosre me badalti, par kabhi 100% transfer nahi hoti.

5. Pollution ka idea (Class 8)

Pollution = environment me harmful cheezon ka milna.

Kuch cheezein khud sad jaati (natural), kuch saalon tak nahi (plastic).

-- END OF CORE BASICS (RED) PAGE --

PART B - CHAPTER KA MAIN CONTENT

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[GREEN = board exam me baar-baar poocha jaata hai]

1. Ecosystem [IMPORTANT]

Ecosystem = ek area ke saare jeev (biotic) + unka nirjeev environment (abiotic) milkar, jo aapas me interact karte.

Eg natural: forest, pond, lake, samudra.

Eg artificial (manav-nirmit): garden, crop field, aquarium.

Components:

- Biotic: Producers, Consumers, Decomposers.
- Abiotic: temperature, light, paani, hawa, mitti, minerals.

2. Producers, Consumers, Decomposers [VERY IMPORTANT]

PRODUCERS (utpadak): khud food banate (green plants, algae) - photosynthesis.

CONSUMERS (upbhokta): doosron ko khaate:

- Herbivores (shakahari): sirf plant khaate (gaay, bakri, deer).
- Carnivores (mansahari): doosre jaanwar khaate (sher, saap).
- Omnivores: dono (insaan, kauwa).
- Parasites: doosre jeev par nirbhar (ju, kide).

DECOMPOSERS (apghatak): mre jeev/waste ko todte (bacteria, fungi).

Importance: nutrients wapas mitti/environment me laate (recycling) -> producers ko phir milte. Inke bina nutrients atak jaate.

3. Food Chain aur Food Web [TOP EXAM AREA]

FOOD CHAIN = energy/food ka ek seedha kram (kaun kisko khata):

Grass -> Deer -> Lion (ya) Plants -> Insect -> Frog -> Snake -> Eagle.

TROPHIC LEVEL = food chain ka har "step" (poshi star):

- T1 = Producers (plants).
- T2 = Primary consumers (herbivores).
- T3 = Secondary consumers (small carnivores).
- T4 = Tertiary consumers (bade carnivores).

FOOD WEB = bahut saari food chains aapas me judi (real nature aise hi hota, ek jeev kai cheezein khata).

4. Flow of Energy aur 10% Law [TOP EXAM - VERY IMP]

Energy ek hi DISHA me behti (unidirectional): Sun -> Producer -> Consumer...

Wapas nahi aati (sun se aati rehti).

10% LAW (Lindeman): ek trophic level se agle level me sirf ~10% energy hi jaati. Baaki 90% respiration/heat/life processes me use ya kho jaati.

Isiliye food chain me aam taur par sirf 3-4 steps hote (energy itni kam ho jaati ki aage support nahi kar sakti).

Energy pyramid: hamesha SEEDHA (upright) - neeche producer (max energy), upar kam.

5. Biological Magnification (Biomagnification) [IMPORTANT]

Harmful chemicals (jaise pesticides/DDT) food chain me upar jaate-jaate JYADA concentrate (jamaa) hote -> isko BIOMAGNIFICATION kehte.

Sabse zyada top consumer me (jaise insaan) - kyunki body se nikalte nahi. Isiliye chemicals/pesticides ka jyada use khatarnak.

6. Biodegradable vs Non-biodegradable [TOP EXAM AREA]

BIODEGRADABLE: jo decomposers (bacteria/fungi) se sad/toot jaate.

Eg: sabzi-chhilke, kaagaz, lakdi, jhoothan, cow dung, cotton.

NON-BIODEGRADABLE: jo aasani se nahi sadte (saalon-saal rehte).

Eg: plastic, polythene, glass, metal, DDT, aluminium can.

Problems:

- Non-biodeg: pollution, mitti/paani kharab, animals nigal lete, naali jaam.
- Biodeg: zyada ho to bhi temporary problem (foul smell, flies) par sad jaata.

7. Ozone Layer aur uska Depletion [VERY IMPORTANT]

Ozone (O₃) = oxygen ke 3 atom; upar atmosphere (stratosphere) me layer.

Kaam: Suraj ki HARMFUL Ultra-Violet (UV) kirno ko rok-ta -> hamein skin cancer, eye (cataract) damage se bachata.

Banta: UV light se O₂ -> O + O ; O + O₂ -> O₃ (upar natural process).

Depletion (patli/chhed hona): CFCs (Chloro-fluoro-carbons - fridge, AC, spray me) ozone ko todte.

1987 MONTREAL PROTOCOL: CFC ka use kam/band karne ka global agreement.

8. Waste Management (kachra prabandhan) [IMPORTANT]

Tareeke: segregation (biodeg/non-biodeg alag), recycling, composting (biodeg ka khaad banana), sewage treatment, landfill, incineration (jalana - control me).

3 R: REDUCE (kam use), REUSE (dobara use), RECYCLE (naya banana).

Garbage ko alag bin: hara (biodegradable wet) + neela (non-biodeg dry).

9. Quick Revision - One Liners

- Ecosystem: biotic + abiotic. Producer/Consumer/Decomposer.
- Food chain (seedha kram); food web (juda hua); trophic level = step.
- Energy: unidirectional, 10% law -> 3-4 steps only; pyramid upright.
- Biomagnification: chemicals top consumer me sabse jyada.
- Biodeg (sabzi/kaagaz) vs non-biodeg (plastic/glass); 3R; ozone UV rokta; CFC ozone todta (Montreal Protocol).

PART C - SOLVED EXAMPLES (HARDEST -> EASIEST)

Example 1 (HARDEST) - 10% law numerical + food chain

Q: Ek food chain: Grass(10000 J) -> Deer -> Lion. 10% law se Deer aur Lion ko kitni energy milegi? Food chain chhoti kyun hoti?

Solution:

- Grass (producer) = 10000 J.
- Deer (T2) ko 10% = 1000 J.
- Lion (T3) ko uska 10% = 100 J.
- Har step par 90% energy heat/respiration me chali jaati -> upar bahut kam bachti -> isliye food chain me sirf 3-4 steps possible.

Example 2 - Biomagnification

Q: DDT (pesticide) food chain me kahan sabse jyada milega aur kyun?

Solution:

- DDT food chain me upar jaate-jaate concentrate hota (biomagnification).
- Top consumer (jaise insaan/eagle) me SABSE JYADA, kyunki yeh body se decompose/nikal nahi paata, har level ka jamaa upar pahunchta.

Example 3 - Decomposers ka mahatva

Q: Agar saare decomposers khatam ho jaayein to kya hoga?

Solution:

- Mre jeev/waste sadenge nahi -> dher lag jaayega.
- Nutrients mitti/environment me wapas nahi aayenge (recycling ruk jaayega).
- Producers ko nutrients nahi milenge -> poora ecosystem disturb.

Example 4 - Biodegradable vs non-biodegradable

Q: In me se kaun biodegradable, kaun non-biodegradable: plastic, sabzi-chhilka, glass, kaagaz?

Solution:

- Biodegradable: sabzi-chhilka, kaagaz (decomposers se sad jaate).
- Non-biodegradable: plastic, glass (aasani se nahi sadte).

Example 5 (EASIEST) - Ozone ka kaam

Q: Ozone layer ka kya kaam hai?

Solution:

- Suraj ki harmful UV kirno ko rokta -> hamein skin cancer/eye damage se bachati.

PART D - SELF TEST (EASY -> HARD)

EASY (1 mark)

- Q1. Biotic aur abiotic me antar (example sahit).
- Q2. Trophic level kya hai?
- Q3. Ozone ka formula + ek kaam.
- Q4. 3 R kaunse hain?

MEDIUM (2-3 marks)

- Q5. Producer, consumer, decomposer ka role.
- Q6. Food chain aur food web me antar.
- Q7. Biodegradable aur non-biodegradable - 3-3 example + 1 problem.
- Q8. Ozone depletion kis se hota + Montreal Protocol.

HARD (3-5 marks, board favourite)

- Q9. 10% law samjhao + numerical (Grass 10000J -> Deer -> Lion).
- Q10. Energy flow unidirectional + food chain 3-4 steps kyun.
- Q11. Biomagnification kya hai + top consumer me sabse jyada kyun.
- Q12. Decomposers khatam ho jaayein to ecosystem par asar.

ANSWER HINTS (PART D)

- A1. Biotic jeevit (paudhe/jaanwar); abiotic nirjeev (hawa/paani/mitti).
- A2. Food chain ka har step/level (producer, consumer...).
- A3. O₃; UV kiranein rokta (skin cancer se bachata).
- A4. Reduce, Reuse, Recycle.
- A5. Producer food banata; consumer khata; decomposer todkar recycle karta.
- A6. Food chain seedha single kram; food web kai chains juda.
- A7. Biodeg: sabzi/kaagaz/lakdi; non-biodeg: plastic/glass/metal; pollution.
- A8. CFC se ozone tootta; Montreal Protocol (1987) CFC band.
- A9. Har level 10% energy; 10000->1000->100 J.
- A10. Energy sun se ek disha; 90% har step kho jaati -> 3-4 step.
- A11. Chemicals upar concentrate; top consumer me jyada (nikalta nahi).
- A12. Waste sadega nahi, nutrients recycle ruk jaayega, ecosystem disturb.

"Padhai tabhi pakki jab khud likh ke test do." - All the best, Krishna!

Generated by Kiro for Krishna1k
